The following sections reproduce the main manuscript Results narrative in full, with each statistical statement expanded to report complete inferential detail as generated from the analysis pipeline: test statistics (F, t, r), degrees of freedom, exact or rounded p values, effect sizes (e.g. Cohen's d, η^2 / partial η^2), and 95% confidence intervals where those quantities are available in the output files. Material is organized in the same RUN order as the comprehensive technical appendix (RUN 1 through RUN 13), so readers can move from this supplement to the corresponding section of that report or to the primary figures and supplementary items cited in the text. Tabular summaries that appear only in the technical appendix are folded into the inline reporting here rather than repeated as separate tables.

RUN 1: Agency did not improve recall performance

For each participant, events were binned according to whether they were remembered or forgotten. Independent raters compared each sentence of recall to the story path read by the participant; if any part of a given event was mentioned in any recall sentence, it was counted as remembered.

There were no significant differences in recall performance across conditions in either story (**Adventure: $F(2,113) = 1.433$, $p = 0.243$, $\eta^2 = 0.0247$, $\eta p^2 = 0.0247$; group means (95% CI): Adventure free: M = 0.543, 95% CI = [0.474, 0.612], n = 22; Adventure pasv: M = 0.474, 95% CI = [0.425, 0.523], n = 49; Adventure yoke: M = 0.503, 95% CI = [0.457, 0.549], n = 45** ; **Romance: $F(2,123) = 0.671$, $p = 0.513$, $\eta^2 = 0.0108$, $\eta p^2 = 0.0108$; group means (95% CI): Romance free: M = 0.455, 95% CI = [0.343, 0.567], n = 18; Romance pasv: M = 0.494, 95% CI = [0.436, 0.552], n = 55; Romance yoke: M = 0.453, 95% CI = [0.410, 0.497], n = 53** ; *Supplementary Figure S3*).

For the Romance story, we additionally recorded the reading time for each subject and measured individual engagement over the course of their reading via a 13-item modified version of the Narrative Transportation scale. There were no significant differences across the three agency conditions on participants' transportation score (**$F(2,123) = 1.818$, $p = 0.167$, $\eta^2 = 0.0287$, $\eta p^2 = 0.0287$; group means (95% CI): Romance free: M = 26.722, 95% CI = [21.286, 32.159], n = 18; Romance pasv: M = 28.855, 95% CI = [26.409, 31.300], n = 55; Romance yoke: M = 31.321, 95% CI = [28.637, 34.005], n = 53**), average reading time per story sentence (**$F(2,123) = 0.405$, $p = 0.668$, $\eta^2 = 0.0065$, $\eta p^2 = 0.0065$; group means (95% CI): Romance free: M = 5.058, 95% CI = [4.149, 5.967], n = 18; Romance pasv: M = 5.381, 95% CI = [4.665, 6.098], n = 55; Romance yoke: M = 4.975, 95% CI = [4.337, 5.613], n = 53**), or the overall reading time for the entire story-path they experienced (**$F(2,123) = 0.423$, $p = 0.656$, $\eta^2 = 0.0068$, $\eta p^2 = 0.0068$; group means (95% CI): Romance free: M = 32.800, 95% CI = [26.911, 38.689], n = 18; Romance pasv: M = 34.879, 95% CI = [30.303, 39.455], n = 55; Romance yoke: M = 32.207, 95% CI = [28.107, 36.307], n = 53**). These results suggest that the overall engagement for the story remained roughly the same across the three agency conditions.

See *Supplement S4* for details about memory for choice and non-choice events; See *Supplement S5* for details about recognition memory performance; See *Supplement S6* for details about memory for denied and granted choice events.

RUN 2: Individual variability in recalled events

Agency magnified individual variability in recall and choice. The Romance story, by design, had half of its events shared across all participants ("shared story sections"), regardless of condition (*Figure 1A*). While participants made many choices during these shared story sections, unbeknownst to them, all choice options led to the same subsequent events. This allowed us to examine inter-participant variability in terms of memory (which events were recalled) and choice behavior (which options were selected) when all events were perfectly matched across participants, i.e., all participants read these events, and the events were composed of identical text.

Individual variability in recalled events. A recall score (0 = Forgotten, 1 = Recalled; see Methods for details) for each of the 64 events in the shared story sections was extracted for each participant, composing a vector of recall performance (*Figure 2A*). To assess the memory similarity across participants, we computed the inter-participant correlation (ISC), i.e., the Pearson correlation between each pair of participants' recall performance vectors, i.e., "Recall ISC".

While events differed in their overall memorability, Recall ISC was significantly above zero in all three conditions (**Romance: Free: mean $r = 0.136$, 95% CI = [0.113, 0.159], $t(152) = 11.746$, $p = 0.000000$, $d = 0.950$** ; **Romance: Yoke: mean $r = 0.226$, 95% CI = [0.218, 0.233], $t(1377) = 58.288$, $p = 0.000000$, $d = 1.570$** ; **Romance: Pasv: mean $r = 0.249$, 95% CI = [0.242, 0.256], $t(1484) = 70.871$, $p = 0.000000$, $d = 1.839$** ; one-sample t-tests against zero), indicating that individuals tended to remember events more similarly to one another than would be expected by chance.

Interestingly, when comparing across the conditions, Free participants had reduced Recall ISC relative to Yoked and Passive participants, indicating that agency induced greater individual variability in terms of which events were recalled (*Figure 2B, left*; **Romance: $F(2,3013) = 48.127$, $p = 0.000000$, $\eta^2 = 0.0310$, $\eta p^2 = 0.0310$** ; post-hoc tests: **Free vs. Passive: $t(1636) = -9.731$, $p = 0.000000$; $d = -0.826$; 95% CI for mean difference = [-0.135, -0.090]** ; **Free vs. Yoked: $t(1529) = -7.285$, $p = 0.000000$; $d = -0.621$; 95% CI for mean difference = [-0.113, -0.065]**).

Out of the 64 events, 15 were "choice events" (the event that the participant chose to occur, e.g., "Sleep beneath the bridge" in *Figure 1B*). To examine whether the reduced Recall ISC observed among Free participants was driven by these choice events, we repeated the analysis using only the 49 non-choice events. The results were largely unchanged: Recall ISC was still significantly above zero in all three conditions (**Romance: Free: mean $r = 0.157$, 95% CI = [0.133, 0.180], $t(152) = 13.212$, $p = 0.000000$, $d = 1.068$** ; **Romance: Yoke: mean $r = 0.219$, 95% CI = [0.210, 0.227], $t(1377) = 51.429$, $p = 0.000000$, $d = 1.385$** ; **Romance: Pasv: mean $r = 0.229$, 95% CI = [0.221, 0.236], $t(1484) = 59.028$, $p = 0.000000$, $d = 1.532$** ; one-sample t-tests against zero), and the Free condition continued to show reduced Recall ISC relative to the Yoked and Passive conditions (*Figure 2C, left*; **Romance: $F(2,3013) = 15.445$, $p = 0.000000$, $\eta^2 = 0.0101$, $\eta p^2 = 0.0101$** ; post-hoc tests: **Free vs. Passive: $t(1636) = -5.679$, $p = 0.000000$; $d = -0.482$; 95% CI for mean difference = [-0.097, -0.047]** ; **Free vs. Yoked: $t(1529) = -4.632$, $p = 0.000004$; $d = -0.395$; 95% CI for mean difference = [-0.088, -0.036]**).

RUN 12: Permutation test for Recall ISC

In the Yoked and Passive conditions, multiple participants followed the story-path corresponding to each of the 18 unique Free participant story-paths. To ensure that the above-reported higher inter-participant memory similarity in the Yoked and Passive conditions was not due to participants sharing the same story-path, we conducted non-parametric tests of Recall ISC (*Figure 2B-C*). We randomly sampled one Yoked and one Passive participant from each of the 18 story-paths to form a sample of 18 Yoked and 18 Passive participants; thus, no pairs within these samples read the same story-path. This process was repeated 10,000 times to generate distributions of Recall ISC for both the Yoked and Passive conditions.

The analysis confirmed that the Free condition's Recall ISC was significantly lower than that of the Passive (**Monte Carlo permutation p = 0.000000 (10,000 resamples; ANALYSIS: 64 Shared Events: Free vs Passive)** ; excluding choice events, **Monte Carlo permutation p = 0.000000 (10,000 resamples; ANALYSIS: 49 Non-Choice Events: Free vs Passive)**) and the Yoked (**Monte Carlo permutation p = 0.000000 (10,000 resamples; ANALYSIS: 64 Shared Events: Free vs Yoked)** ; excluding choice events **Monte Carlo permutation p = 0.007000 (10,000 resamples; ANALYSIS: 49 Non-Choice Events: Free vs Yoked)**) conditions (*Figure 2B-C, right*).

RUN 3: Individual variability in choices made

The option that was selected (1 or 2) at each of the 15 choice-points in the shared story sections was extracted for each Free and Yoked participant (Passive participants made no choices), composing a vector of choice selections. To assess the choice similarity across participants, we computed the inter-participant (Pearson) correlation between each pair of participants' choice selection vectors, i.e., "Choice ISC".

Within-group Choice ISC was significantly above zero in both conditions (**Free: mean r = 0.208, 95% CI = [0.165, 0.251], t(152) = 9.548, p = 0.000000, d = 0.772** ; **Yoke: mean r = 0.255, 95% CI = [0.241, 0.269], t(1377) = 36.193, p = 0.000000, d = 0.975**), showing that certain choice options were intrinsically preferred over others. Comparing across conditions, Free participants had significantly reduced Choice ISC compared to Yoked participants (**two-sample t-test: t = -2.105, p = 0.035; d = -0.179; 95% CI for mean difference = [-0.091, -0.003]**), indicating that agency (full as opposed to partial) induced greater individual variability in terms of which options were selected.

RUN 13: Permutation test with matching Choice ISC

This increased Choice ISC within the Free condition did not drive their increased Recall ISC. Another permutation test showed that Free participants still had significantly reduced within-group Recall ISC compared to their Yoked counterparts with matching Choice ISC (**Monte Carlo p = 0.000000 (64 shared events; 10,000 valid resamples; matching Choice ISC); 49 non-choice: p = 0.015400**); see *Supplement S12*.

These results suggest that agency was the driver of greater individual variability in both memory and choice behaviors.

RUN 4: Divergence from the group

Each Free participant's memory divergence score was calculated as one minus the Pearson correlation between their recall performance vector and the group averaged recall performance vector. In other words, the more different their memory performance vector was from the group average, the higher their memory divergence score. Similarly, we calculated each Free participant's choice divergence score as one minus the Pearson correlation between their choice selection vector and the group averaged choice selection vector.

Memory divergence and choice divergence were correlated with each other (**Romance: r(16) = 0.405, 95% CI for r = [-0.076, 0.733], p = 0.095 (N = 18)** ; **r(98) = 0.296, 95% CI for r = [0.106, 0.465], p = 0.0028 (N = 100)**). In other words, the more idiosyncratic their memory for shared events, the more idiosyncratic their choices.

Overall, these results support the idea that agency magnified individual variability in, i.e., personalized, both memory and choice behaviors. This effect was observed while all events were held constant across conditions.

RUN 5: Event recall was predicted by semantic and causal centrality

Narrative networks were computed for each unique story path following the methods of Lee & Chen. For semantic narrative network analysis, each event was converted into an embedding vector using the Universal Sentence Encoder (USE). Semantic centrality, a measure of how strongly interconnected a given event was with other events in the narrative via shared meaning, was calculated for each event by averaging its embedding cosine similarity with all other events in the story path. The effect of semantic centrality (semantic influence) on memory was computed as the Pearson correlation between semantic centrality and recall (an event-by-event vector of remembered=1, forgotten=0) for each participant.

For both stories, semantic centrality significantly predicted recall, i.e., a significant semantic influence on memory was observed, in all three conditions (**Adventure FREE Semantic: mean r = 0.223, 95% CI = [0.140, 0.306], t(21) = 5.582, p = 0.000015, d = 1.190; Adventure YOKE Semantic: mean r = 0.323, 95% CI = [0.282, 0.363], t(44) = 16.071, p = 0.000000, d = 2.396; Adventure PASV Semantic: mean r = 0.308, 95% CI = [0.259, 0.356], t(48) = 12.681, p = 0.000000, d = 1.812; Romance FREE Semantic: mean r = 0.146, 95% CI = [0.096, 0.197], t(17) = 6.109, p = 0.000012, d = 1.440; Romance YOKE Semantic: mean r = 0.221, 95% CI = [0.198, 0.245], t(52) = 19.021, p = 0.000000, d = 2.613; Romance PASV Semantic: mean r = 0.265, 95% CI = [0.239, 0.292], t(54) = 20.003, p = 0.000000, d = 2.697** ; *Supplement S8* anc *Supplementary Figure S8-2*).

For causal narrative network analysis, independent human raters judged which pairs of events were causally linked in a given story path (Adventure: 1 rater per path; Romance: average of 3 raters per path; see Methods). Causal centrality, a measure of an event's causal connectedness to other events within a narrative, was calculated for each event by averaging across its causal connections with all other events in the story path. The effect of causal centrality (causal influence) on memory was computed as the Pearson correlation between causal centrality and event-by-event recall for each participant.

For both stories, causal centrality significantly predicted recall, i.e., a significant causal influence on memory was observed, in all three conditions (**Adventure FREE Causal: mean r = 0.187, 95% CI = [0.102, 0.271], t(21) = 4.604, p = 0.000153, d = 0.982; Adventure YOKE Causal: mean r = 0.159, 95% CI = [0.100, 0.217], t(44) = 5.453, p = 0.000002, d = 0.813; Adventure PASV Causal: mean r = 0.126, 95% CI = [0.075, 0.178], t(48) = 4.922, p = 0.000011, d = 0.703; Romance FREE Causal: mean r = 0.213, 95% CI = [0.161, 0.266], t(17) = 8.613, p = 0.000000, d = 2.030; Romance YOKE Causal: mean r = 0.236, 95% CI = [0.203, 0.269], t(52) = 14.398, p = 0.000000, d = 1.978; Romance PASV Causal: mean r = 0.244, 95% CI = [0.224, 0.265], t(54) = 23.988, p = 0.000000, d = 3.235** ; *Supplement S8* and *Supplementary Figure S8-2*).

In sum, both semantic centrality and causal centrality predicted recall of interactive narratives, echoing the findings of earlier studies on the effects of semantic and causal relations on memory for narratives.

RUN 6: Agency reduced the influence of semantic but not causal centrality on recall

We next compared the strength of semantic and causal influences on memory across the three conditions (Free, Yoked, Passive). Importantly, because Yoked and Passive participants read the story paths generated by Free participants, event content was matched across conditions; only the degree of perceived agency varied.

For both stories, we observed significant differences across conditions, wherein Free had lower semantic influence on memory compared to Yoked and Passive (**Adventure: $F(2,113) = 3.036$, $p = 0.052$, $\eta^2 = 0.0510$, $\eta p^2 = 0.0510$; post-hoc: free vs yoke: $t(65) = -2.503$, $p = 0.015$, $d = -0.612$; free vs pasv: $t(69) = -1.889$, $p = 0.063$, $d = -0.448$** ; **Romance: $F(2,123) = 11.455$, $p = 0.000027$, $\eta^2 = 0.1570$, $\eta p^2 = 0.1570$; post-hoc: free vs yoke: $t(69) = -3.090$, $p = 0.0029$, $d = -0.733$; free vs pasv: $t(71) = -4.429$, $p = 0.000034$, $d = -1.037$; yoke vs pasv: $t(106) = -2.493$, $p = 0.014$, $d = -0.480$** ; *Figure 3A*). In contrast, causal influence on memory was not different between conditions (*Figure 3B*). There was a significant network type x agency interaction for the Adventure story (**$F(2,113) = 3.272$, $p = 0.042$, $\eta^2 = 0.0547$, $\eta p^2 = 0.0547$ (network type × agency interaction)**) and a trend for the same for the Romance story (**$F(2,123) = 2.990$, $p = 0.054$, $\eta^2 = 0.0464$, $\eta p^2 = 0.0464$ (network type × agency interaction)**).

These results show that when participants had agentive control over the plot of a narrative, semantic influence (the impact of semantic centrality) on later memory was reduced; meanwhile, causal influence (the impact of the causal centrality) on recall was relatively unaffected. This weakening of semantic influence suggests that Free participants' semantic space may have shifted away from the generic (normative) semantic space captured by generic text embeddings. In contrast, causal influence was unchanged by agency.

RUN 7: Agency introduces temporal dependencies in memory

We examined whether recall performance for a given event could be predicted by whether its temporally neighboring events at encoding were recalled, which we term the "neighbor encoding effect". First, for each participant and for each event, we calculated the average of the recall scores for the immediately previous and next events at encoding (the neighbors); for the first and last event, there were neighbors on only one side, and thus these entries consisted merely of recall performance for the next and previous event, respectively. This procedure generated a vector of neighbor recall performance for each participant. We then calculated the neighbor encoding effect as the correlation between the neighbor recall performance vector and the original recall performance vector for each participant (*Figure 4A*).

The neighbor encoding effect was positive in all three conditions for both stories (**Adventure: FREE: mean neighbor encoding $r = 0.282$, $t(21) = 6.103$, $p = 0.000005$; YOKE: mean neighbor encoding $r = 0.235$, $t(44) = 6.193$, $p = 0.000000$; PASV: mean neighbor encoding $r = 0.239$, $t(48) = 8.714$, $p = 0.000000$ Romance: FREE: mean neighbor encoding $r = 0.352$, $t(17) = 7.589$, $p = 0.000001$; YOKE: mean neighbor encoding $r = 0.186$, $t(52) = 7.409$, $p = 0.000000$; PASV: mean neighbor encoding $r = 0.126$, $t(54) = 6.426$, $p = 0.000000$**), and significantly different across the three conditions in the Romance story, with Free having a higher neighbor encoding effect compared to Yoked and Passive (**$F(2,123) = 12.073$, $p = 0.000016$, $\eta^2 = 0.1641$, $\eta p^2 = 0.1641$** ; post-hoc tests: **Free vs. Yoked: $t(69) = 3.255$, $p = 0.0018$, $d = 0.773$ Free vs. Passive: $t(71) = 5.238$, $p = 0.000002$, $d = 1.226$** ; *Figure 4B*). Note that despite having the same trend, the Adventure story did not show a statistically significant agency enhancement. This is because the Passive participants reached a ceiling for the neighbor encoding effect due to limited and vastly varying story length (Adventure vs. Romance: 22-59 events vs. 128-135 events). However, additional analysis confirmed that agency can enhance neighbor encoding effect in longer Adventure stories, supporting the result for the Romance story; see *Supplement S13*.

Overall, these results suggest that agency enhanced the tendency for temporally neighboring events at encoding to share the same subsequent memory status, either both recalled or both forgotten.

RUN 8: Temporal violation rate

The neighbor encoding effect is distinct from the "temporal contiguity effect", which describes the phenomenon that recalling one item from a randomized list tends to trigger the recall of items which were experienced nearby in time; the neighbor encoding effect does not incorporate any information about the temporal order of recall. To examine temporal order effects during recall in our data, we calculated the temporal violation rate as the frequency with which each participant recalled events out of order.

For each participant, recall was divided into segments (brief sentences). We counted the number of times that a recall segment referred to an event that occurred earlier in the story than the events referred to by the previous recall segment; this was then divided by the participant's total number of recall segments. There was no significant difference in temporal violation rate across conditions in either story (**Adventure: $F(2,113) = 2.564$, $p = 0.081$, $\eta^2 = 0.0434$, $\eta p^2 = 0.0434$** ; **Romance: $F(2,123) = 0.096$, $p = 0.908$, $\eta^2 = 0.0016$, $\eta p^2 = 0.0016$**). In general, temporal order was remarkably well-preserved, with low temporal violation rates in all conditions (*Figure 4C-D*).

RUN 9: Greater memory divergence was associated with weaker semantic influence

We next examined how memory divergence scores (*Figure 2*) were related to a) the impact of semantic centrality on recall (*Figure 3*) and b) the neighbor encoding effect (*Figure 4*). Each Free participant's semantic influence score was obtained by calculating the Pearson correlation between semantic centrality and memory performance (same as shown in *Figure 3*).

Memory divergence was negatively correlated with semantic influence scores in Free participants. In other words, the more a participant's recall deviated from other participants in the group, the weaker the effect of semantic centrality on their recall.

This was true when including only the 18 Free participants who had yoked counterparts (**Romance: $r(16) = -0.519$, 95% CI for $r = [-0.794, -0.069]$, $p = 0.027$ ($N = 18$)**) as well as when using the full sample (**$r(98) = -0.460$, 95% CI for $r = [-0.602, -0.290]$, $p = 0.000001$ ($N = 100$)** ; *Supplement S10* and *Supplementary Figure S10A*). Choice divergence, however, was not significantly correlated with semantic network scores (**Romance: $r(16) = -0.111$, 95% CI for $r = [-0.549, 0.376]$, $p = 0.662$ ($N = 18$)** ; **$r(98) = -0.164$, 95% CI for $r = [-0.349, 0.034]$, $p = 0.104$ ($N = 100$)**). Note that these comparisons could only be made for the Romance story, as the analyses of divergence depend on the shared story sections.

RUN 10: Neighbor encoding effect correlations

The neighbor encoding effect was also positively associated with memory divergence in Free participants (**Romance: $r(16) = 0.496$, 95% CI for $r = [0.038, 0.782]$, $p = 0.036$ ($N = 18$)** ; **$r(98) = 0.304$, 95% CI for $r = [0.114, 0.472]$, $p = 0.0021$ ($N = 100$)** ; *Supplementary Figure S10B*). In other words, the more a participant's recall deviated from other participants in the group, the more that participant's neighboring events tended to have the same recall status (remembered or forgotten).

The neighbor encoding effect was negatively correlated with semantic influence scores in Free participants, across both stories (**Adventure: $r(20) = -0.348$, 95% CI for $r = [-0.671, 0.087]$, $p = 0.113$ ($N = 22$)** . **Romance: $r(16) = -0.324$, 95% CI for $r = [-0.687, 0.168]$, $p = 0.189$ ($N = 18$)** ; **$r(98) = -0.347$, 95% CI for $r = [-0.509, -0.162]$, $p = 0.000402$ ($N = 100$)** ; *Supplementary Figure S10C*). However, when including both semantic influence and the neighbor encoding scores in a multiple linear regression predicting memory divergence, semantic influence score was a significant predictor (**Romance (N=100): semantic influence β = -0.415, p = 0.000047; model R^2 = 0.235**) while the neighbor encoding effect only showed a trenc (**Romance (N=100): neighbor encoding β = 0.089, p = 0.087**).

Overall, the degree to which a participant's memory was idiosyncratic in terms of which events they recalled (memory divergence) was negatively correlated with impact of the story's semantic network on memory – in line with the idea that agency personalizes memory.

RUN 11: Consequences of having your choices denied

In the results reported above, Yoked subjects generally exhibited similar memory performance to the Passive subjects—similar idiosyncrasy in event recall, in semantic and causal centrality effect on memory, and in neighbor encoding effects, with their group mean falling in between that of the Free and Passive condition. Given that the Yoked condition had a varied amount of choice granted/denied (see *Supplementary Figure S6-1*), these results aligned with the design of 'partial agency' for the Yoked condition in a gradient of agency.

Nonetheless, having one's agency denied can have unique memory effects at local choice events: one's memory for the denied choice events is selectively reduced compared to its choice-granted counterparts in the Free condition

(**Adventure: t(88) = -2.494, p = 0.015, d = -0.526, 95% CI for mean difference = [-0.253, -0.029]** ; **Romance: t(104) = -2.437, p = 0.017, d = -0.473, 95% CI for mean difference = [-0.186, -0.019]** , two-sample t-test); individual differences contribute to a variation of tendency to selective recall or forget the denied choice events (see *Supplement S6* for details).

The percentage of choices granted in the Yoked participants was not predictive of individual's recall performance, recall similarity to their Free and Passive condition counterparts, semantic and causal centrality effects on memory, nor their neighbor encoding effects (all ps>.3); however, higher percentage of choices granted predicted greater individual tendency to forget the choice-denied events (PE-boost (correlation between want-not and recall vectors per subject) vs percentage wanted: **Adventure: r(42) = 0.335, 95% CI for r = [0.042, 0.575], p = 0.026 (N = 44); PE-boost vs. % choices wanted** ; **Romance: r(49) = 0.137, 95% CI for r = [-0.144, 0.398], p = 0.336 (N = 51); PE-boost vs. % choices wanted** ; see *Supplement S7* for details).

Together, these results suggest that in a context lacking full agentive control, perceived agency and their effects on memory could vary across individuals in non-systematic ways. The one exception is that with more control in such agency-uncertain contexts, the more one has reduced recall for the agency-denied events.